ELEN 635 Electromagnetics-Notes set #1

- Purpose: The purpose of this course is to teach you some of the basic properties of high frequency electromagnetic fields and circuit analysis techniques
- Major areas/application
  a) Scattering and Propagation
    - Wireless device coverage area
    - Radar
    - EMI (Interference), EMC (Compatibility)

  b) Antennas
    - Wired and Wireless devices
    - Autos, aircrafts
    - Magnetic resonance

  c) High frequency circuits
    - Microwave analog circuits (5G, 6G)
    - Computer high pulse rate digital circuits
    - Nanotechnology

  d) Optics
    - Circuits
    - Reflection an refraction

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$
$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega\varepsilon\mathbf{E}$$
$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\varepsilon}$$
$$\nabla \cdot \mathbf{H} = 0$$

Helmholtz equation
$$\nabla^2\mathbf{E} + k^2\mathbf{E} = 0$$

Solution
$$\mathbf{E} = \mathbf{E}_0 e^{-jk\hat{\mathbf{k}}\cdot\mathbf{r}}$$
$$\mathbf{E}_0 = E_x\hat{\mathbf{x}} + E_y\hat{\mathbf{y}} + E_z\hat{\mathbf{z}}$$
$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

Propagation constant or Wave number
$$k = \frac{2\pi}{\lambda} = \frac{\omega}{v_p}$$
$$v_p = 1/\sqrt{\mu\varepsilon}$$

Add Loss σ (S/m) $= 1/\rho$
- $k \to \gamma = \alpha + j\beta$
- $E_x = E_0 e^{-\gamma z} = E_0 e^{-\alpha z} e^{-j\beta z}$
- Skin depth $\delta_d = 1/\alpha$
- Loss tangent $= \sigma/\omega\varepsilon$

$$\boldsymbol{\mathcal{E}}(\mathbf{r},t) = \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega)e^{j\omega t}\right\}$$
example
$$= \mathrm{Re}\left\{(E_0 e^{-jkz}\hat{\mathbf{x}})e^{j\omega t}\right\}$$
$$= E_0\cos(\omega t - kz)\hat{\mathbf{x}}$$

Wave Polarization
$$\mathbf{E} = \underbrace{E_{01}e^{-jkz}}_{E_x}\hat{\mathbf{x}} + \underbrace{E_{02}e^{-j(kz+\theta)}}_{E_y}\hat{\mathbf{y}}$$
- linear polarization $E_{01} = E_{02} = E_0,\ \theta = 0°$
- Right or left hand circular $E_{01} = E_{02} = E_0,\ \theta = \mp 90°$
- Right or left hand elliptical $E_{01} \neq E_{02}$ or $\theta = \mp 90°$

Boundary Conditions
$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$$
$$\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$
$$\hat{\mathbf{n}} \cdot (\varepsilon_1\mathbf{E}_1 - \varepsilon_2\mathbf{E}_2) = \rho_s$$
$$\hat{\mathbf{n}} \cdot (\mu_1\mathbf{H}_1 - \mu_2\mathbf{H}_2) = 0$$

- Fields reflected Γ and transmitted T through a boundary
- Snell's law
- Critical angle
- Brewster angle
- Penetration depth

TL equations
$$\frac{dV}{dz} = (R + j\omega L)I$$
$$\frac{dI}{dz} = (G + j\omega C)V$$

Helmholtz equations
$$\frac{dV^2}{dz^2} - \gamma^2 V = 0$$
$$\frac{dI^2}{dz^2} - \gamma^2 I = 0$$

Solutions
$$V(z) = V_0 e^{-j\beta z}\left(1 + \Gamma_L e^{-j2\beta z}\right)$$
$$I(z) = \frac{V_0}{Z_0} e^{-j\beta z}\left(1 - \Gamma_L e^{-j2\beta z}\right)$$

$$S = \frac{1+|\Gamma|}{1-|\Gamma|}, \quad \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$
$$Z_{in}(l) = Z_0 \frac{Z_L + jZ_0\tan(\beta l)}{Z_0 + jZ_L\tan(\beta l)}$$

In the beginning God created the heavens and the earth. And the earth was without form, and void, and darkness was on the face of the deep. And the spirit of God moved upon the face of the waters. And God said:

'Maxwell's equations in time domain point (differential) form'

$$\nabla \times \mathcal{E}(\mathbf{r},t) = -\frac{\partial}{\partial t}\mathcal{B}(\mathbf{r},t) - \mathcal{M}(\mathbf{r},t) \tag{1}$$

$$\nabla \times \mathcal{H}(\mathbf{r},t) = \frac{\partial}{\partial t}\mathcal{D}(\mathbf{r},t) + \mathcal{J}(\mathbf{r},t) \tag{2}$$

$$\nabla \cdot \mathcal{D}(\mathbf{r},t) = \rho(\mathbf{r},t) \tag{3}$$

$$\nabla \cdot \mathcal{B}(\mathbf{r},t) = \rho_m(\mathbf{r},t) \tag{4}$$

and there was light.

Take the divergence of (2)

$$\nabla \cdot \left[\nabla \times \mathcal{H}(\mathbf{r},t)\right] = \frac{\partial}{\partial t}\nabla \cdot \mathcal{D}(\mathbf{r},t) + \nabla \cdot \mathcal{J}(\mathbf{r},t)$$

vector identity   from (3)

$$0 = \frac{\partial}{\partial t}\rho(\mathbf{r},t) + \nabla \cdot \mathcal{J}(\mathbf{r},t)$$

$$\Rightarrow \quad -\frac{\partial}{\partial t}\rho(\mathbf{r},t) = \nabla \cdot \mathcal{J}(\mathbf{r},t) \tag{5}$$

This is known as the *equation of continuity* which is a <u>conservation law</u> of electromagnetics.

All accepted theories of physics are expected to have a continuity equation. Electromagnetics has several but (5) is the most important.

A **conversion between the time and frequency** (time harmonic) domains (see Appendix B of these notes), would typically be written:

$$\boldsymbol{\mathcal{E}} = \mathrm{Re}(\mathbf{E}e^{j\omega t}) = \mathrm{Re}(|\mathbf{E}|e^{j\alpha}e^{j\omega t}) = |\mathbf{E}|\cos(\omega t + \alpha)$$

Converting Maxwell's eqns. to time harmonic form with ' Re ' notation

Example: $\nabla \times \boldsymbol{\mathcal{E}}(t) = -\frac{\partial}{\partial t}\boldsymbol{\mathcal{B}}(t) - \boldsymbol{\mathcal{M}}(t)$

$$\nabla \times \mathrm{Re}\left\{\mathbf{E}(\omega)\, e^{j\omega t}\right\} = -\frac{\partial}{\partial t}\mathrm{Re}\left\{\mathbf{B}(\omega)\, e^{j\omega t}\right\} - \mathrm{Re}\left\{\mathbf{M}(\omega)\, e^{j\omega t}\right\}$$

$$\mathrm{Re}\left\{\nabla \times \mathbf{E}(\omega)\, e^{j\omega t}\right\} = -\mathrm{Re}\left\{\mathbf{B}(\omega)\frac{\partial}{\partial t} e^{j\omega t}\right\} - \mathrm{Re}\left\{\mathbf{M}(\omega)\, e^{j\omega t}\right\}$$

$$\mathrm{Re}\left\{\nabla \times \mathbf{E}(\omega)\, e^{j\omega t}\right\} = -\mathrm{Re}\left\{j\omega\mathbf{B}(\omega) e^{j\omega t}\right\} - \mathrm{Re}\left\{\mathbf{M}(\omega)\, e^{j\omega t}\right\}$$

$$\mathrm{Re}\left\{\left[\nabla \times \mathbf{E}(\omega) + j\omega\mathbf{B}(\omega) + \mathbf{M}(\omega)\right] e^{j\omega t}\right\} = 0$$

This can only be true for all times t if $[\quad] = 0$

$$\Rightarrow \nabla \times \mathbf{E} = -j\omega\mathbf{B} - \mathbf{M} \qquad \text{(1a)}$$

Similarly

$$\nabla \times \mathbf{H} = j\omega\mathbf{D} + \mathbf{J} \qquad \text{(2a)}$$

$$\nabla \cdot \mathbf{D} = \rho \qquad \text{(3a)}$$

$$\nabla \cdot \mathbf{B} = \rho_m \qquad \text{(4a)}$$

$$\nabla \cdot \mathbf{J} = -j\omega\rho \qquad \text{(5a)}$$

which are Maxwell's equations and the equation of continuity in frequency domain point form.

- Converting from point (differential) form to integral form:

Stokes theorem

$$\int_s (\nabla \times \mathbf{F}) \cdot d\mathbf{s} = \oint \mathbf{F} \cdot d\mathbf{l}$$

Divergence theorem

$$\int_\upsilon \nabla \cdot \mathbf{F} dv = \oint_s \mathbf{F} \cdot d\mathbf{s}$$

Example 1: Convert $\nabla \times \mathbf{E} = -j\omega \mathbf{B} - \mathbf{M}$ to integral form.

Integrate both sides

$$\int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = \int_s (-j\omega \mathbf{B} - \mathbf{M}) \cdot d\mathbf{s}$$

Apply Strokes theorem

$$\oint \mathbf{E} \cdot d\mathbf{l} = -j\omega \int_s \mathbf{B} \cdot d\mathbf{s} - \underbrace{\int_s \mathbf{M} \cdot d\mathbf{s}}_{=\mathrm{I_m}}$$

The integral form of $\nabla \times \mathbf{H} = j\omega \mathbf{D} + \mathbf{J}$ is

$$\oint \mathbf{H} \cdot d\mathbf{l} = j\omega \int \mathbf{D} \cdot d\mathbf{s} + \underbrace{\int \mathbf{J} \cdot d\mathbf{s}}_{=\mathrm{I}}$$

Example 2: Convert $\nabla \cdot \mathbf{D} = \rho$ to integral form

Integrate both sides over a volume v

$$\int_v \nabla \cdot \mathbf{D} dv = \int_v \rho dv$$

Apply divergence theorem

$$\oint \mathbf{D} \cdot d\mathbf{s} = q$$

The integral form of $\nabla \cdot \mathbf{B} = \rho_m$ is

$$\oint \mathbf{B} \cdot d\mathbf{s} = q_m$$

Most of our work in time harmonic electromagnetics is in the point (differential equation) form, equations (1)-(5).

Incorporating material properties into electromagnetic equations

Definitions $\varepsilon \doteq$ permittivity $\left(F/m\right)$

$$\varepsilon_r = \varepsilon/\varepsilon_0 \doteq \text{dielectric constant (relative permittivity)}$$

$$\mu \doteq \text{ permeability } \left(H/m\right)$$

$$\mu_r = \mu/\mu_0 \doteq \text{ relative permeability}$$

$$\sigma \doteq \text{conductivity } \left(\frac{S}{m}\right)$$

- Common types of mediums:

  Nondispersive - μ, ε are not a function of frequency (otherwise dispersive)
  Homogeneous - μ, ε are not a function of position (otherwise inhomogeneous)
  Linear - μ, ε not a function of field amplitude (otherwise nonlinear)
  Isotropic - μ, ε not a function of direction (otherwise anisotropic)

- Example: nondispersive, homogeneous, linear, isotropic medium <u>constitutive relations</u>:

$$\left.\begin{aligned}\mathbf{D} &= \varepsilon\mathbf{E}\\ \mathbf{B} &= \mu\mathbf{H}\end{aligned}\right\} \Rightarrow \quad D_x = \varepsilon E_x$$

Maxwell's equations become:

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} - \mathbf{M} \qquad \text{(1c)}$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E} + \mathbf{J} \qquad \text{(2c)}$$

$$\nabla \cdot \mathbf{E} = \rho/\varepsilon \qquad \text{(3c)}$$

$$\nabla \cdot \mathbf{H} = \rho_m/\mu \qquad \text{(4c)}$$

- Example: Anisotropic crystal

$$\mathbf{D} = \overline{\boldsymbol{\varepsilon}} \cdot \mathbf{E}$$

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \Rightarrow D_x = \varepsilon_{xx}E_x + \varepsilon_{xy}E_y + \varepsilon_{xz}E_z$$

$\overline{\boldsymbol{\varepsilon}}$ here is a 'dielectric tensor'.

Constitutive parameters and relations for linear media

$$\mathbf{D} = \dot{\varepsilon}\mathbf{E}$$

$$
\begin{aligned}
\nabla \times \mathbf{H} &= \mathbf{J} + j\omega \mathbf{D} \\
&= \mathbf{J}^i + \mathbf{J}^c + \mathbf{J}^d \\
&= \mathbf{J}^i + \sigma\mathbf{E} + j\omega\dot{\varepsilon}\mathbf{E} \\
&= \mathbf{J}^i + \underbrace{\left(\sigma + \omega\varepsilon'' + j\omega\varepsilon'\right)}_{\doteq j\omega\varepsilon_c}\mathbf{E}
\end{aligned}
$$

$\mathbf{J}^i$ − current impressed by a source generator

$\mathbf{J}^c$ − conduction current ($= \sigma\mathbf{E}$ Ohm's law)

$\mathbf{J}^d$ − displacement current

$\dot{\varepsilon}$ $= \varepsilon' - j\varepsilon''$ - medium complex permittivity

$$= \mathbf{J}^i + \varepsilon_c\mathbf{E} \;\; ; \;\; \varepsilon_c = \sigma + \omega\varepsilon'' + j\omega\varepsilon'$$

(eqn. 2-66 in Balanis)

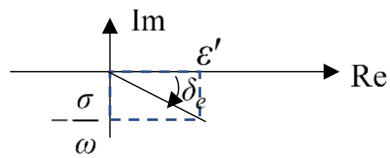
conduction losses

Polarization and/or ionic losses

electric energy Storage in a dielectric material ie polarization or ionic delay in time

$$
\begin{aligned}
\nabla \times \mathbf{H} &= \mathbf{J}^i + (j\omega\dot{\varepsilon} + \sigma)\mathbf{E} \\
&= \mathbf{J}^i + j\omega\varepsilon'(1 - j\frac{\varepsilon''}{\varepsilon'} - j\frac{\sigma}{\omega\varepsilon'})\mathbf{E} = \mathbf{J}^i + j\omega\varepsilon'(1 - j\frac{\sigma}{\omega\varepsilon'})\mathbf{E} \\
&= \mathbf{J}^i + j\omega\varepsilon'(1 - j\tan\delta_e)\mathbf{E}
\end{aligned}
$$

loss tangent Balanis pg 69



See Balanis Table 2.5 for typical dielectric constants and loss tangents

The quality factor Q of a material

$$Q = \frac{\textit{maximum stored energy}}{\textit{average power loss}} \cdot \omega = \frac{\varepsilon'}{\varepsilon''} = \frac{1}{\tan\delta_a}$$

See Appendix F of these notes for the derivation of this equation

Similarly for a magnetic material

$$
\begin{aligned}
\nabla \times \mathbf{E} &= -\mathbf{M}^i - j\omega\mathbf{B} \\
&= -\mathbf{M}^i - j\omega\left(\mu' - j\mu''\right)\mathbf{H} \\
&= -\mathbf{M}^i - j\omega\dot{\mu}\mathbf{H}
\end{aligned}
$$

$$\dot{\mu} = \mu'' + j\mu'$$

known impressed magnetic current

magnetization loss

magnetic energy storage in a magnetic material

- Vector Helmholtz equation

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} - \mathbf{M}^i$$

$$\nabla \times \nabla \times \mathbf{E} = -j\omega\mu\nabla \times \mathbf{H} - \nabla \times \mathbf{M}^i$$

vector identity

M.E.'s

$$-\nabla^2\mathbf{E} + \nabla(\nabla \cdot \mathbf{E}) = -j\omega\mu[\mathbf{J}^i + (\sigma + j\omega\varepsilon)\mathbf{E}] - \nabla \times \mathbf{M}^i$$

$$-\nabla^2\mathbf{E} = -j\omega\mu\mathbf{J}^i + \underbrace{\left(\omega^2\mu\varepsilon - j\omega\mu\sigma\right)}_{\doteq k_s^2}\mathbf{E} \underbrace{-\nabla\left(\nabla \cdot \mathbf{E}\right)} - \nabla \times \mathbf{M}^i$$

$$\nabla^2\mathbf{E} + k_s^2\mathbf{E} = j\omega\mu\mathbf{J}^i - \frac{\nabla\left(\nabla \cdot \mathbf{J}^i\right)}{\sigma + j\omega\varepsilon} + \nabla \times \mathbf{M}^i \qquad \text{Helmholtz Equation}$$

where:

$$\nabla \times \mathbf{H} = \mathbf{J}^i + \sigma\mathbf{E} + j\omega\varepsilon\mathbf{E}$$

$$\underbrace{\nabla \cdot \left(\nabla \times \mathbf{H}\right)}_{=0} = \nabla \cdot \mathbf{J}^i + \nabla \cdot \left(\sigma + j\omega\varepsilon\right)\mathbf{E}$$

$$\nabla \cdot \mathbf{E} = \frac{-\nabla \cdot \mathbf{J}^i}{\sigma + j\omega\varepsilon}$$

- Special case – Helmholtz equation in a source free region, $\mathbf{J}^i = \mathbf{M}^i = 0$

$$\nabla^2\mathbf{E} + k_s^2\mathbf{E} = 0$$

or decomposed becomes

$$\left.\begin{aligned} \nabla^2 E_x + k_s^2 E_x &= 0 \\ \nabla^2 E_y + k_s^2 E_y &= 0 \\ \nabla^2 E_z + k_s^2 E_z &= 0 \end{aligned}\right\} \text{3 independent scalar Helmholtz eqns.}$$

or

$$\frac{\partial^2}{\partial x^2}E_x + \frac{\partial^2}{\partial y^2}E_x + \frac{\partial^2}{\partial z^2}E_x + k_s^2 E_x = 0$$

$$\frac{\partial^2}{\partial x^2}E_y + \frac{\partial^2}{\partial y^2}E_y + \frac{\partial^2}{\partial z^2}E_y + k^2 E_y = 0$$

$$\frac{\partial^2}{\partial x^2}E_z + \frac{\partial^2}{\partial y^2}E_z + \frac{\partial^2}{\partial z^2}E_z + k^2 E_z = 0$$

Example A uniform (no variation transverse to propagation direction) plane wave (no field component in propagation direction) is propagating in the z-direction and polarized in the x-direction.

$$\frac{d^2}{dz^2}E_x + k_s{}^2 E_x = 0$$

$$E_x = E_x^+ e^{-jk_s z} + E_x^- e^{jk_s z}$$

when $\varepsilon_s = \varepsilon - j\sigma/\omega$ and $k_s = -j\gamma$ where $\gamma = \alpha + j\beta$ in Balanis

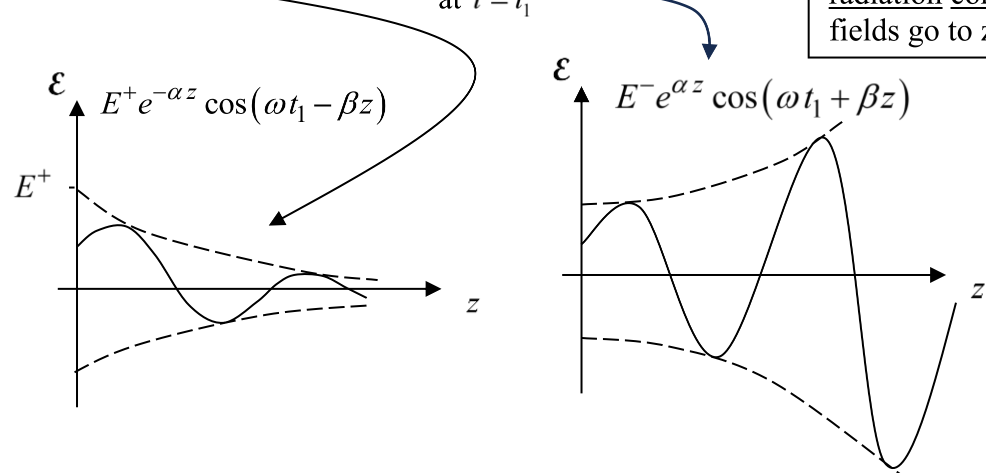
$$E_x = E^+ e^{-\alpha z} e^{-j\beta z} + E^- e^{\alpha z} e^{+j\beta z}$$

In the time domain this becomes

$$\mathcal{E}_x(t) = \text{Re}\left\{E_x e^{j\omega t}\right\}$$
$$= \underbrace{E^+ e^{-\alpha z}\cos(\omega t - \beta z)} + \underbrace{E^- e^{\alpha z}\cos(\omega t + \beta z)}$$

at $t = t_1$

This solution eliminated by the Sommerfeld radiation condition: all fields go to zero at



Skin depth $\delta_d \doteq$ the depth at which the field has decayed to $\frac{1}{e} = e^{-1}$ of its initial value:

$$e^{-\alpha\delta_d} = e^{-1} \Rightarrow \delta_d = \frac{1}{\alpha}$$

The exponential decay of a wave never goes to zero until it reaches infinity so skin depth is used to provide a relative number for how rapidly the field decays with distance.

- How do we know that this wave is traveling in the $+z$ direction?

  Ans: If an observer and a wave are moving at the same velocity, the observer would always see the same amplitude when he looks over at the wave. So

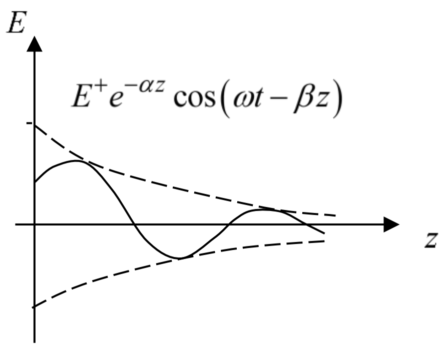
  $E_o \cos(\omega t - \beta z) = a$ (a constant)

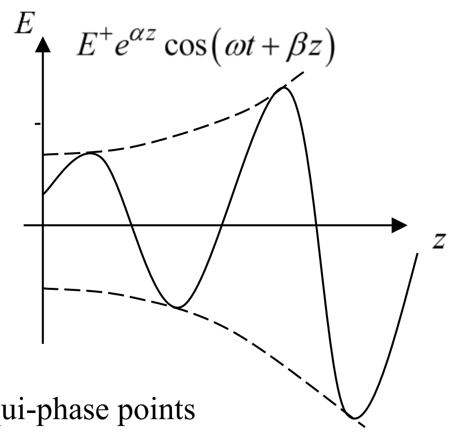
  $\Rightarrow \omega t - \beta z = b$ (a constant)

  $$\omega - \beta \frac{dz}{dt} = 0 \Rightarrow \frac{dz}{dt} \doteq \upsilon_p = \frac{\omega}{\beta}$$

  Since ω and β are both positive the phase velocity is positive, so this wave is traveling in the $+z$ direction.

- For γ we could have chosen either a + or a – sign.. A plus sign would mean the field traveling in the negative z direction.





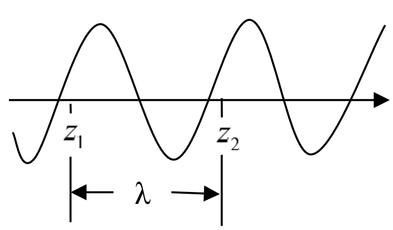
- Wavelength $\doteq$ the distance between two equi-phase points

  Example: For the plane wave $\boldsymbol{\mathcal{E}} = E_o \cos(\omega t - \beta z)$ the phase difference between 2 equi-phase points z_1 and z_2 is 2π

$$(\omega t - \beta z_2) - (\omega t - \beta z_1) = 2\pi$$
$$\beta(z_1 - z_2) = 2\pi$$

$$|z_1 - z_2| = 2\pi/\beta \doteq \lambda$$

or $\beta = \dfrac{2\pi}{\lambda}$



When $\varepsilon_s = \varepsilon - j\sigma/\omega$

$$k \to k_s = \omega\sqrt{\mu\varepsilon_s} = \sqrt{-j\omega\mu(\sigma + j\omega\varepsilon)} \doteq -j\gamma$$
$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \doteq \alpha + j\beta$$

$$\Rightarrow \beta = \omega\sqrt{\mu\varepsilon}\left[\tfrac{1}{2}\left[\sqrt{1+\left(\sigma/\omega\varepsilon\right)^2} + 1\right]\right]^{1/2}$$
$$\alpha = \omega\sqrt{\mu\varepsilon}\left[\tfrac{1}{2}\left[\sqrt{1+\left(\sigma/\omega\varepsilon\right)^2} - 1\right]\right]^{1/2}$$

- From above

$$\upsilon_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon}\sqrt{\frac{1}{2}\left[\sqrt{1+\left(\sigma/\omega\varepsilon\right)^2} + 1\right]}}$$
$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{\mu\varepsilon}\sqrt{\frac{1}{2}\left[\sqrt{1+\left(\sigma/\omega\varepsilon\right)^2} + 1\right]}}$$

Notice that phase velocity and wavelength depend on frequency, permittivity and <u>conductivity.</u>

<u>Case 1</u> no conductivity $\sigma = 0$

$$k = \omega\sqrt{\mu\varepsilon}$$
$$\upsilon_\rho = \frac{1}{\sqrt{\mu\varepsilon}}$$
$$\lambda = \frac{2\pi}{\omega\sqrt{\mu\varepsilon}}$$

<u>Case 2</u> good conductor $\dfrac{\sigma}{\omega\varepsilon} \gg 1$

$$\beta = \alpha = \sqrt{\frac{\omega\mu\sigma}{2}}$$
$$\Rightarrow \upsilon_\rho = \sqrt{\frac{2\omega}{\mu\sigma}}$$
$$\lambda = 2\pi\sqrt{\frac{2}{\omega\mu\sigma}}$$

<u>Case 3</u> perfect conductor $\sigma = \infty$

$$\left.\begin{aligned} \beta = \alpha &= \infty \\ \upsilon_\rho &= 0 \\ \lambda &= 0 \end{aligned}\right\} \Rightarrow \text{ no fields in perfect conductor when } \omega \neq 0.$$

<u>Power</u>

$$\left.\begin{array}{l}\nabla\times\mathbf{H}=\mathbf{J}^t\\ \nabla\times\mathbf{E}=-\mathbf{M}^t\end{array}\right\}\xrightarrow[\text{Dot with }\ \mathbf{H}^*]{\text{Conjugate and dot with }\mathbf{E}}\begin{array}{l}\mathbf{E}\cdot\left(\nabla\times\mathbf{H}^*=\mathbf{J}^{t*}\right)\\ \mathbf{H}^*\cdot\left(\nabla\times\mathbf{E}=-\mathbf{M}^t\right)\end{array}$$

Subtract

$$\underbrace{\mathbf{E}\cdot\left(\nabla\times\mathbf{H}^*\right)-\mathbf{H}^*\cdot\left(\nabla\times\mathbf{E}\right)}_{\downarrow}=\mathbf{E}\cdot\mathbf{J}^{t*}+\mathbf{H}^*\cdot\mathbf{M}^t$$

vector identity
(Appendix II)

$$\downarrow$$

$$-\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right)\ =\mathbf{E}\cdot\mathbf{J}^{t^*}+\mathbf{H}^*\cdot\mathbf{M}^t$$

Expand $\mathbf{J}^t$ and $\mathbf{M}^t$

$$\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right)+\mathbf{E}\cdot\left[\left(\sigma+j\omega\varepsilon\right)^*\mathbf{E}^*+\mathbf{J}^{i*}\right]+\mathbf{H}^*\cdot\left[j\omega\mu\mathbf{H}+\mathbf{M}^i\right]=0$$

$$\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right)+\sigma E^2-j\omega\varepsilon E^2+\mathbf{E}\cdot\mathbf{J}^{i*}+j\omega\mu H^2+\mathbf{H}^*\cdot\mathbf{M}^i=0$$

Rearranging so that the source (excitation) terms are on the right hand side

$$\underbrace{\nabla\cdot\underbrace{\left(\mathbf{E}\times\mathbf{H}^*\right)}_{\doteq\mathbf{S}}}_{\substack{P_f\ =\ \text{complex}\\ \text{vol. power density}\\ \text{leaving a point}}}+\underset{\substack{\doteq P_d\\ \text{dissipated}\\ \text{vol.}\\ \text{power}\\ \text{density}}}{\sigma E^2}+j2\omega\underbrace{(\underbrace{-\tfrac{1}{2}\varepsilon E^2}_{-w_e}+\underbrace{\tfrac{1}{2}\mu H^2}_{w_m})}_{\substack{\text{stored electric}\\ \text{and magnetic}\\ \text{energy}\\ \text{density}}}=\underbrace{-\mathbf{E}\cdot\mathbf{J}^{i*}-\mathbf{H}^*\cdot\mathbf{M}^i}_{\substack{\text{power}\\ \text{density}\\ \text{supplied}\\ \text{by the}\\ \text{sources}}}$$

<u>Examples</u> A plane wave, $\mathbf{E}=E_0e^{-jkz}\hat{x}$, $\mathbf{H}=E_0/\eta_0e^{-jkz}\hat{y}$, where $\eta_0=\sqrt{\mu_0/\varepsilon_0}$, and $\mathbf{J}^i=\mathbf{M}^i=0$ in air.

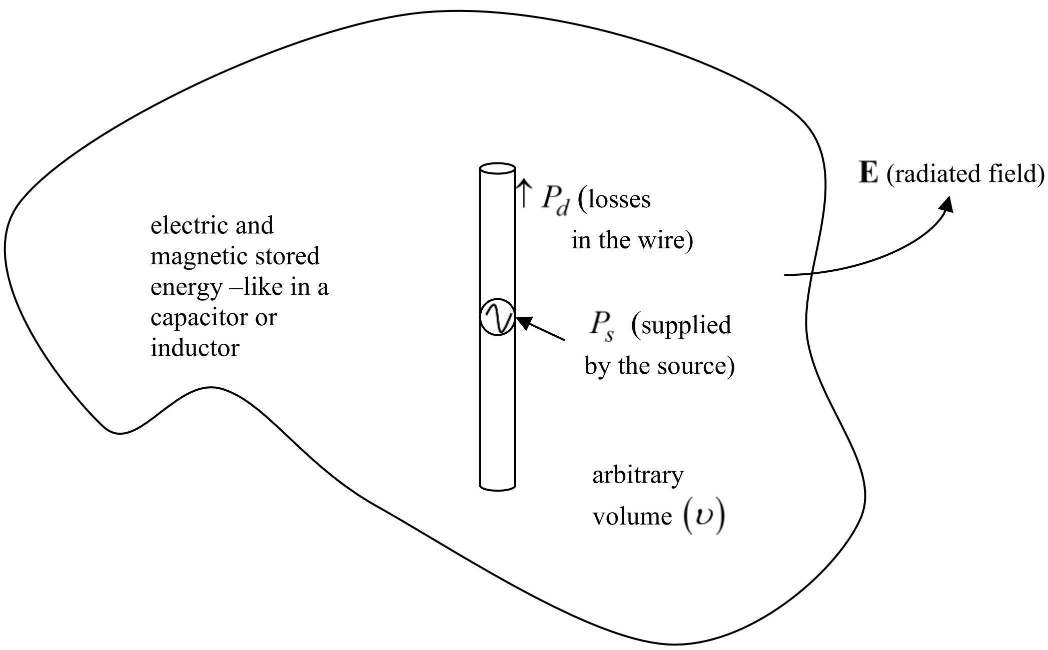
1) <u>Net</u> time ave. power density passing through a point:
   $\mathrm{Re}\left\{\mathbf{E}\times\mathbf{H}^*\right\}=E_0^2/\eta_0\ \hat{z}$
2) <u>Net</u> volume power density loss at a point:
   $\nabla\cdot\mathrm{Re}\left\{\mathbf{E}\times\mathbf{H}^*\right\}=-\sigma E^2=0$ $\qquad$ ($\sigma=0$ in air)
3) <u>Net</u> stored volume power density at a point:
   $\nabla\cdot\mathrm{Im}\left\{\mathbf{E}\times\mathbf{H}^*\right\}=-j2\omega\left(-\tfrac{1}{2}\varepsilon_0E^2+\tfrac{1}{2}\mu_0H^2\right)=-j2\omega\left[-\tfrac{\varepsilon_0}{2}E_0^2+\tfrac{\mu_0}{2}\left(\tfrac{E_0^2}{\eta_0^2}\right)\right]=0$

- Integrate over a volume to get the time average quantities (Yariv pg 84)

$$\underbrace{\int\left[\nabla\cdot\left(\mathbf{E}\times\mathbf{H}^*\right)\right]d\upsilon}_{=\oint\left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s}\doteq P_f} + \underbrace{\int \sigma E^2 d\upsilon}_{\doteq P_d} + j2\omega\underbrace{\left\{-\frac{1}{2}\int\varepsilon E^2 d\upsilon + \frac{1}{2}\underbrace{\int \mu H^2 d\upsilon}_{W_m}\right\}}_{W_e} + \underbrace{\int\left(\mathbf{E}\cdot\mathbf{J}^{i*}+\mathbf{H}\cdot\mathbf{M}^i\right)d\upsilon}_{\doteq P_s} = 0$$

power entering the region through the surface	power dissipated	net electric energy stored	net magnetic energy stored	power supplied in the region

<u>example</u> – dipole antenna



- When $J=0$, $M=0$, which is the case when the volume υ does not include the source, then

power dissipated $-\mathrm{Re}\left\{\oint\left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s}\right\} = \int \sigma E^2 d\upsilon$

power stored $\quad -\mathrm{Im}\left\{\oint\left(\mathbf{E}\times\mathbf{H}^*\right)\cdot d\mathbf{s}\right\} = 2\omega\left\{-\frac{1}{2}\int_{vol}\varepsilon E^2 d\upsilon + \frac{1}{2}\int_{vol}\mu H^2 d\upsilon\right\}$

Time average power density $\doteq$ $\quad \boldsymbol{\mathcal{S}}_{av} = \dfrac{1}{T}\displaystyle\int_0^T \boldsymbol{\mathcal{E}}(t)\times\boldsymbol{\mathcal{H}}(t)dt = \dfrac{1}{T}\int_0^T \boldsymbol{\mathcal{S}}\ dt$

$\boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{E}}(t)\times\boldsymbol{\mathcal{H}}(t) \doteq$ instantaneous Poynting vector

Example: plane wave in free space

$$\boldsymbol{\mathcal{E}} = E_0\cos(\omega t+\phi)\hat{x}$$

$$\boldsymbol{\mathcal{H}} = H_0\cos(\omega t+\theta)\hat{y}$$

Method 1

$$\boldsymbol{\mathcal{S}} = E_0H_0\cos(\omega t+\phi)\cos(\omega t+\theta)\,\overset{\hat{z}}{\hat{x}\times\hat{y}}$$

$$= \frac{1}{2}E_0H_0\left[\cos\left(\phi-\theta\right)+\cos\underbrace{\left(2\omega t+\phi+\theta\right)}\right]\hat{z}$$

$$\left[\left(\omega t+\phi\right)-\left(\omega t+\theta\right)\right] \qquad \left[\left(\omega t+\phi\right)+\left(\omega t+\theta\right)\right]$$

$$\boldsymbol{\mathcal{S}}_{av} = \frac{1}{2T}\int_0^T\left[E_0H_0\cos\left(\phi-\theta\right)+\overset{0}{E_0H_0\cos\left(2\omega t+\phi+\theta\right)}\right]dt\,\hat{z}$$

$$\left(\text{use } \int_0^T\cos(a+bx)dx = \frac{1}{b}\sin(a+b)\ \ and\ \ \omega = 2\pi f = \frac{2\pi}{T}\right)$$

$$= \frac{1}{2}E_0H_0\cos(\phi-\theta)\,\hat{z}$$

Method 2

$$\mathbf{E} = E_0e^{j\phi}\hat{x}$$

$$\mathbf{H} = H_0e^{j\theta}\hat{y}$$

$$\boldsymbol{\mathcal{S}}_{av} = \frac{1}{2}\mathrm{Re}\left\{\mathbf{E}\times\mathbf{H}^*\right\} = \frac{1}{2}\mathrm{Re}\left\{E_0e^{j\phi}\hat{x}\times H_0e^{-j\theta}\hat{y}\right\} = \frac{1}{2}E_0H_0\cos(\phi-\theta)\,\hat{z}$$

- Since we are usually doing calculations in the frequency domain we would prefer to be able to immediately calculate the time average power without changing back to the time domain. The result above shows that we <u>can</u> get the time average power using frequency domain quantities:

$$\boldsymbol{\mathcal{S}}_{av} = \frac{1}{T}\int_0^T\boldsymbol{\mathcal{S}}dt = \frac{1}{2}\mathrm{Re}\left\{\mathbf{S}(\omega)\right\}$$

$$= \frac{1}{T}\int_0^T\left(\mathbf{E}(t)\times\mathbf{H}(t)\right)dt = \frac{1}{2}\mathrm{Re}\left\{\mathbf{E}\times\mathbf{H}^*\right\}$$

δ - function

$$\int_a^b \delta(x-c)\,dx = 1 \qquad ; \qquad a \le c \le b$$

$$\int_a^b \delta(x-c) f(x)\,dx = f(c) \quad , \quad a \le c \le b$$

- Cylindrical coordinates: $dv = \rho d\phi\, d\rho\, dz$

$$\int_a^b \delta(\rho-\rho')\,d\rho = 1 \qquad ; \qquad a \le \rho' \le b$$

$$\int_a^b \left[\frac{\delta(\phi-\phi')}{\rho}\right] \rho d\phi = 1 \qquad ; \qquad a \le \phi' \le b$$

$$\int_a^b \delta(z-z')\,dz = 1 \qquad ; \qquad a \le z' \le b$$

- Spherical coordinates: $dv = r^2 \sin\theta dr d\theta d\phi$

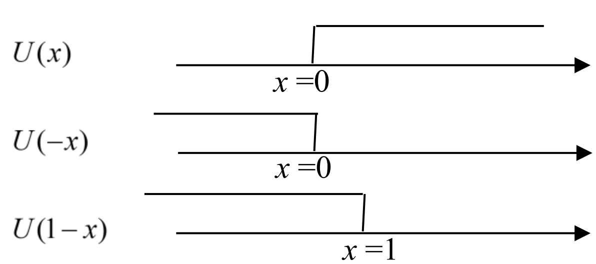
$$= \underbrace{r\sin\theta d\phi}\, r d\theta\, dr$$

$$\int_a^b \left[\frac{\delta(\phi-\phi')}{r\sin\theta}\right] r\sin\theta\, d\phi = 1 \qquad ; \qquad a \le \phi' \le b$$

$$\int_a^b \left[\frac{\delta(\theta-\theta')}{r}\right] r d\theta = 1 \qquad ; \qquad a \le \theta' \le b$$

$$\int_a^b \delta(r-r')\,dr = 1 \qquad ; \qquad a \le r' \le b$$

Unit Step Function: Zero except when the argument is positive



Appendix A: RMS Quantities

As an example: The effective value of a time varying current is equal to the value of the direct current which, flowing through a resistance R, delivers the same power to R as the periodic current does. The average power delivered to the resistor by the periodic current $i(t)$ having a period T is

$$P = \frac{1}{T}\int_0^T i^2 R dt = \frac{1}{T}\int_0^T i^2 dt\, R \tag{1A}$$

The power delivered by a direct current I is related to the power delivered by time varying current by creating an effective current I_{eff} ,

$$P = I^2 R = I_{eff}^2 R \tag{2A}$$

Equating the above two expressions gives

$$I_{eff} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} \doteq I_{RMS} \tag{3A}$$

According to Eqn. (3A) the effective current is the square root of the mean of the square of the time varying current. For this reason the effective current is referred to as the root mean square or RMS current.

The most important special case is that of the sinusoidal current

$$i(t) = \mathrm{Re}(Ie^{j\omega t}) = \mathrm{Re}(I_m e^{j\theta} e^{j\omega t}) = I_m \cos(\omega t - \theta) \tag{4A}$$

which when placed in (3A) gives

$$I_{RMS} = \frac{I_m}{\sqrt{2}} \tag{5A}$$

The RMS current is real, independent of phase angle, and equal to the maximum of the sinusoidal current divided by the square root of two. So when time average power is calculated using a sinsusoidal current the equation is usually

$$P_{AVE} = \mathrm{Re}(I_{RMS}^2 R) = \frac{1}{2}\mathrm{Re}(I_m^2 R) \tag{6A}$$

Appendix B

What is the meaning of

$$\mathcal{E}(\mathbf{r},t) = \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega)e^{j\omega t}\right\}? \tag{1B}$$

Recall the Fourier transform ($\mathcal{F}$) pair

$$F(\omega) = \int_{-\infty}^{\infty} F(t)e^{-j\omega t}dt \doteq \mathcal{F}\{F(t)\} \tag{2B}$$

$$F(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega \doteq \mathcal{F}^{-1}\{F(\omega)\} \tag{3B}$$

from which one gets:

$$\delta(\omega-\omega') = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-j(\omega-\omega')t}dt \tag{4B}$$

Set

$$\mathcal{E}(\mathbf{r},t) = \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega)e^{j\omega t}\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \underbrace{\chi(\mathbf{r},\omega)}_{\substack{\text{frequency}\\ \text{spectrum of}\\ \mathcal{E}(\mathbf{r},t)}} e^{j\omega t}d\omega \tag{5B}$$

Take the Fourier transform of both sides

$$\int_{-\infty}^{\infty} \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega')e^{j\omega' t}\right\}e^{-j\omega t}dt = \chi(\mathbf{r},\omega) \tag{6B}$$

Using an equivalent expression for $\mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega')e^{j\omega' t}\right\}$

$$\begin{gathered}\int_{-\infty}^{\infty} \left\{\frac{\mathbf{E}(\mathbf{r},\omega')e^{j\omega' t} + \mathbf{E}^*(\mathbf{r},\omega')e^{-j\omega' t}}{2}\right\}e^{-j\omega t}dt = \chi(\mathbf{r},\omega) \\ \frac{\mathbf{E}(\mathbf{r},\omega')}{2}\int_{-\infty}^{\infty} e^{-j(\omega-\omega')t}dt + \frac{\mathbf{E}^*(\mathbf{r},\omega')}{2}\int_{-\infty}^{\infty} e^{-j(\omega+\omega')t}dt = \chi(\mathbf{r},\omega)\end{gathered} \tag{7B}$$

$\pi\mathbf{E}(\mathbf{r},\omega')\delta(\omega-\omega') + \pi\mathbf{E}^*(\mathbf{r},\omega')\delta(\omega+\omega') = \chi(\mathbf{r},\omega)$ ($\doteq \mathbf{E}(\mathbf{r},\omega)$ in RWP King)

$\therefore$ For time harmonic fields

$$\mathcal{E}(\mathbf{r},t) = \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega')e^{j\omega' t}\right\} = \mathcal{F}^{-1}\left\{\pi\mathbf{E}(\mathbf{r},\omega')\delta(\omega-\omega') + \pi\mathbf{E}^*(\mathbf{r},\omega')\delta(\omega+\omega')\right\}$$

or equivalently

$$\mathcal{E}(\mathbf{r},t) = \mathrm{Re}\left\{\mathbf{E}(\mathbf{r},\omega)e^{j\omega t}\right\} = \mathcal{F}^{-1}\left\{\pi\mathbf{E}(\mathbf{r},\omega)\delta(\omega-\omega') + \pi\mathbf{E}^*(\mathbf{r},\omega)\delta(\omega+\omega')\right\} \tag{8B}$$

Where the inverse Fourier transform is taken over ω' coordinates

Note: $\mathbf{E}^*(\mathbf{r},\omega) = \mathbf{E}(\mathbf{r},-\omega)$

Appendix C

Time harmonic case comparison of Re$\{\ \}$ and Fourier Transform ($\mathcal{F}\{\ \}$) methods

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)G(\mathbf{r},t) = -\delta(\mathbf{r}-\mathbf{r}')\cos(\omega t) \tag{1C}$$

<u>Using Fourier Transform method -</u> $\mathcal{F}$(1C)

$$\left(\nabla^2 + k'^2\right)G(\mathbf{r},\omega') = -\delta(\mathbf{r}-\mathbf{r}')\pi\left[\delta(\omega-\omega')+\delta(\omega+\omega')\right] \tag{2C}$$

The solution to (2C) is

$$G(\mathbf{r},\omega') = \frac{e^{-jk'r}}{4\pi r}\pi\left[\delta(\omega-\omega')+\delta(\omega+\omega')\right] \tag{3C}$$

Inverse Fourier Transform - $\mathcal{F}^{-1}$ (3C)

$$\begin{aligned} G(\mathbf{r},t) &= \frac{\pi}{4\pi r}\frac{1}{2\pi}\int_{-\infty}^{\infty}\left\{e^{j\omega'\left(\frac{r}{c}-t\right)}\delta(\omega-\omega') + e^{-j\omega'\left(\frac{r}{c}-t\right)}\delta(\omega+\omega')\right\}d\omega' \\ &= \frac{\cos(\omega t - kr)}{4\pi r} \end{aligned} \tag{4C}$$

<u>Now using Re notation method</u> – Start by reverse engineering (1C)

$$\begin{aligned} &\nabla^2\,\mathrm{Re}\left\{G(\mathbf{r},\omega)e^{j\omega t}\right\} - \frac{1}{c^2}\mathrm{Re}\left\{(j\omega)^2 G(\mathbf{r},\omega)e^{j\omega t}\right\} + \delta(\mathbf{r}-\mathbf{r}')\mathrm{Re}\left\{e^{j\omega t}\right\} = 0 \\ &\mathrm{Re}\left\{\left[\nabla^2 G(\mathbf{r},\omega) - \frac{1}{c^2}(j\omega)^2 G(\mathbf{r},\omega) + \delta(\mathbf{r}-\mathbf{r}')\right]e^{j\omega t}\right\} = 0 \end{aligned} \tag{5C}$$

$$\Rightarrow \left(\nabla^2 + k^2\right)G(\mathbf{r},\omega) = -\delta(\mathbf{r}-\mathbf{r}') \tag{6C}$$

- note (6C) $\neq$ (2C)

The solution to (6C) is

$$G(\mathbf{r},\omega) = \frac{e^{-jkr}}{4\pi r} \tag{7C}$$

- note (7C) $\neq$ (3C)

Converting to the time domain

$$\begin{aligned} G(\mathbf{r},t) &= \mathrm{Re}\left\{\frac{e^{-jkr}}{4\pi r}e^{j\omega t}\right\} \\ &= \frac{\cos(\omega t - kr)}{4\pi r} \end{aligned} \tag{8C}$$

- Note (8C) = (4C)

$\therefore$ Fourier Transform and Re methods give different intermediate results but yield the same time domain results with less work.

Appendix D

Fourier Transform solution method for the wave equation

$$\left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)G(\mathbf{r},t) = -\delta(\mathbf{r}-\mathbf{r}')\delta(t-t') \tag{1D}$$

Define $\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt$ (2D)

Fourier transformed Eqn. (1D) becomes

$$\left(\nabla^2 + k^2\right)G(\mathbf{r},\omega) = -\delta(\mathbf{r}-\mathbf{r}')e^{-j\omega t'} \tag{3D}$$

The solution to (3D) is

$$G(\mathbf{r},\omega) = G(\mathbf{r})e^{-j\omega t'} \tag{4D}$$
$$= \frac{e^{-jkr}}{4\pi r}e^{-j\omega t'}$$

Define $\mathcal{F}^{-1}\{f(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\omega)e^{j\omega t}d\omega$ (5D)

Inverse Fourier transformed, Eqn. (4D) becomes (with $k = \omega/c$)

$$G(\mathbf{r},t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{e^{-j\omega(t'-t+r/c)}}{4\pi r}d\omega \tag{6D}$$
$$= \frac{\delta\left(t'-[t-r/c]\right)}{4\pi r}$$

Appendix E

Electrical Properties of materials – dispersion and nonlinearity

Electric susceptibility models the polarization response of a dielectric due to an applied electric field. The polarization response of a material is proportional to the electric susceptibility. A number of properties of the dielectric are therefore wrapped up in the susceptibility function. Or it could also be said that the susceptibility provides the relationship between the electric field $\mathbf{E}$ and the magnetic flux density $\mathbf{D}$,

$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} = \varepsilon_0\mathbf{E} + \varepsilon_0\chi_e\mathbf{E} = \varepsilon_0(1+\chi_e)\mathbf{E} = \varepsilon_0\varepsilon_r\mathbf{E} = \varepsilon\mathbf{E}$ (2-11) Balanis pg. 46

ε_r table pg. 47 at 'static frequencies'

So $1+\chi_e = \varepsilon_r$ or $\chi_e = \varepsilon_r - 1$.

But ε_r is a constant only if χ_e is a constant. There can be many different types of molecular polarization in materials, so to account for all of the types of polarization we write,

$$\begin{aligned}\mathbf{P} &= \mathbf{P}_0 + \mathbf{P}^{(1)} + \mathbf{P}^{(2)} + \mathbf{P}^{(3)} + \ldots \\ &= \varepsilon_0 \chi^{(0)} \mathbf{E} + \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(2)} \mathbf{E}^2 + \varepsilon_0 \chi^{(3)} \mathbf{E}^3 + \ldots\end{aligned}$$

The flux density, with only the first polarization present, is

$$\begin{aligned}\mathbf{D}_0 &= \varepsilon_0 \mathbf{E} + \mathbf{P}_0 \\ &= \varepsilon_0 \mathbf{E} + \varepsilon_0 \chi^{(0)} \mathbf{E} \\ &= \varepsilon_0 \left(1 + \chi^{(0)}\right) \mathbf{E}\end{aligned}$$

where $\chi^{(0)}$ refers to linear, non-dispersive susceptibility.

$\mathbf{P}_0 = 0$ for materials where the polarization is zero, which is the case for ferroelectric materials. In other cases at relatively low frequencies (in the gigahertz range) the definition $1 + \chi^{(0)} \doteq \varepsilon_\infty$ is used to account for a non-unity permittivity, giving

$$\mathbf{D}_0 = \varepsilon_0 \mathbf{E} + \varepsilon_0 \chi^{(0)} \mathbf{E} = \varepsilon_0 \left(1 + \chi^{(0)}\right) \mathbf{E} = \varepsilon_0 \varepsilon_\infty \mathbf{E}$$

and therefore adding in the other susceptibility terms

$$\begin{aligned}\mathbf{D} &= \varepsilon_0 + \varepsilon_0 \chi^{(0)} \mathbf{E} + \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(2)} \mathbf{E}^2 + \varepsilon_0 \chi^{(3)} \mathbf{E}^3 + \ldots \\ &= \varepsilon_0 + \varepsilon_0 \left(\chi^{(0)} + \chi^{(1)} + \chi^{(2)} \mathbf{E} + \chi^{(3)} \mathbf{E}^2 + \ldots\right) \mathbf{E}\end{aligned}$$

$$\mathbf{D} = \varepsilon_0 (1 + \chi^{(0)}) \mathbf{E} + \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(2)} \mathbf{E}^2 + \varepsilon_0 \chi^{(3)} \mathbf{E}^3 + \ldots$$

or

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E} + \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(2)} \mathbf{E}^2 + \varepsilon_0 \chi^{(3)} \mathbf{E}^3 + \ldots$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E} + \mathbf{P}^{(1)} + \mathbf{P}^{(2)} + \mathbf{P}^{(3)} + \ldots \doteq \varepsilon \mathbf{E} + \mathbf{P}$$

$\chi^{(1)}$ is a linear term that <u>accounts for dispersion</u> and therefore depends upon frequency and is commonly expressed in terms of a Lorentz dispersion formula.

$\chi^{(2)}$ and $\chi^{(3)}$ are known as the second- and third-order optical susceptibilities which <u>account for 2nd and 3rd order nonlinearities</u> respectively. However, the processes that account for these nonlinearities are very different. So $\mathbf{P}^{(2)}$ and $\mathbf{P}^{(3)}$ are 2nd and 3rd order nonlinear polarizations [Boyd].

$\chi^{(2)}$ can only occur in certain types of crystals (noncentrosymmetric crystals), but these are rare.

$\chi^{(3)}$ is a non-linear term that accounts for Ramon and Kerr nonlinearities.

The current density is obtained by taking the time derivative of the polarization,

$$\begin{aligned}\frac{\partial \mathbf{D}}{\partial t} &= \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \frac{\partial \mathbf{P}^{(1)}}{\partial t} + \frac{\partial \mathbf{P}^{(2)}}{\partial t} + \frac{\partial \mathbf{P}^{(3)}}{\partial t} + \ldots \\ &= \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{J}_3 + \ldots\end{aligned}$$

Which in the frequency domain is

$$j\omega \mathbf{D} = j\omega\varepsilon \mathbf{E} + \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{J}_3 + \ldots$$

All of the terms on the right hand side are the result of polarization responses of the material to the applied field. The first term on the right hand side contains what is described as 'static permittivity'. The word 'static' has to do with its relatively constant value over a wide frequency range, usually microwave and below.

Ampere's law in Maxwell's equations in the frequency domain becomes

$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega \mathbf{D} = \mathbf{J}_i + \mathbf{J}_c + j\omega\varepsilon \mathbf{E} + \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{J}_3 + \ldots$$

The first term on the right hand side is the 'impressed' current, due to the generator. The effects described in Balanis Chapter 2 are all linear, so they refer only to the second and third terms on the right hand side in the equation above. Conduction loss, which involves σ and is not a polarization effect, is accounted for in the second term on the right where $\mathbf{J}_c = \sigma_s \mathbf{E}$, Ohm's law. The subscript s in σ_s indicates that this conductivity is measured in with static DC current.

Now, ignoring the nonlinear terms and accounting only for the source $\mathbf{J}_i$, dispersion $\mathbf{J}_1$, and loss $\mathbf{J}_c$:

$$\begin{aligned}\nabla \times \mathbf{H} &= \mathbf{J}_i + j\omega\varepsilon \mathbf{E} + \mathbf{J}_c + \mathbf{J}_1 \\ &= \mathbf{J}_i + j\omega\varepsilon_0(1+\chi^{(0)})\mathbf{E} + \sigma_s \mathbf{E} + j\omega\varepsilon_0\chi^{(1)}\mathbf{E} \\ &= \mathbf{J}_i + j\omega\varepsilon_0(1+\chi^{(0)}+\chi^{(1)})\mathbf{E} + \sigma_s \mathbf{E} \\ &= \mathbf{J}_i + j\omega\varepsilon_0\dot{\varepsilon}_r \mathbf{E} + \sigma_s \mathbf{E} \\ &= \mathbf{J}_i + j\omega\dot{\varepsilon}\mathbf{E} + \sigma_s \mathbf{E} \\ &= \mathbf{J}_i + j\omega\left(\dot{\varepsilon} - j\frac{\sigma_s}{\omega}\right)\mathbf{E}\end{aligned}$$

where $\dot{\varepsilon}_r = 1 + \chi^{(0)} + \chi^{(1)}$. The static and nonlinear terms, $\chi^{(0)}$ and $\chi^{(1)}$, can both be complex so define $\dot{\varepsilon} = \varepsilon' + j\varepsilon"$. This expression inserted into the equation above gives

$$\begin{aligned}\nabla \times \mathbf{H} &= \mathbf{J}_i + (\sigma_s + \omega\varepsilon")\mathbf{E} + j\omega\varepsilon'\mathbf{E} \\ &= \mathbf{J}_i + \sigma_e \mathbf{E} + j\omega\varepsilon'\mathbf{E}\end{aligned}$$

where $\sigma_e \doteq \sigma_s + \omega\varepsilon"$. This is the Maxwell-Ampere equation. Now adding in the nonlinear terms

$$\nabla \times \mathbf{H} = \mathbf{J}_i + \sigma_e \mathbf{E} + j\omega\varepsilon' \mathbf{E} + \mathbf{J}_2 + \mathbf{J}_3 + \ldots$$

Balanis, Advanced Engineering Electromagnetics book gives the values of ε' and $\varepsilon"$ on pages 72 and 73. The nonlinear terms are very complicated and tedious to calculate but can be found in some books like R.W. Boyd, Nonlinear Optics.

There are some simple formulas for linear complex permittivity. However, these require that the real part of the complex permittivity must be known at DC and at very high, ideally infinite, frequency. Balanis gives the formula for Debye media, which is a medium that can be characterized with a single pole (resonance). In research concerning high power electromagnetic waves propagating in the air it is popular to characterize the atmosphere as a Lorentzian medium. A Lorentzian model has one or more pairs of complex conjugate poles. For a single complex conjugate set of poles this formula is:

$$\dot{\varepsilon}_r = \varepsilon_r' - j\varepsilon_r'' = \varepsilon_{r\infty} + \frac{(\varepsilon_{rs}' - \varepsilon_{r\infty})\omega_p^2}{\omega_p^2 - 2j\omega\gamma_p - \omega^2}$$

Where ω_p is the frequency of the pole pair and γ_p is the damping coefficient, both of which can be obtained from material measurement data.

There seems to be some discrepancy in the value of ε_∞ in the literature. Frankly we have found that its use is mainly as a 'fudge factor' to make the dielectric function $\dot{\varepsilon}$ fit the measured data at high frequency. Because the measurements do not always agree, neither do ε_∞'s .

See Yariv Quantum Electronics pg 87 for anisotropic crystal materials where the polarization (**P**) and the electric field are not in the same direction.

$$\mathbf{D} = \bar{\boldsymbol{\varepsilon}} \cdot \mathbf{E}$$

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix} \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \Rightarrow D_x = \varepsilon_{xx}E_x + \varepsilon_{xy}E_y + \varepsilon_{xz}E_z$$

Here $\bar{\boldsymbol{\varepsilon}}$ is a 'dielectric tensor'.

Appendix F The quality factor of a material

$$\nabla\times\mathbf{H}=\mathbf{J}^{i}+(j\omega\dot{\varepsilon}+\sigma)\mathbf{E}$$

$$=\mathbf{J}^{i}+j\omega\varepsilon'(1-j\frac{\varepsilon''}{\varepsilon'}-j\frac{\sigma}{\omega\varepsilon'})\mathbf{E}=\mathbf{J}^{i}+j\omega\varepsilon'(1-j\tan\delta_a-j\tan\delta_s)\mathbf{E}$$ loss tangent Balanis pg 74

$$=\mathbf{J}^{i}+j\omega\varepsilon'(1-j\tan\delta_e)\mathbf{E}$$

$$\gamma^2=\left[j\omega\mu(\sigma+j\omega\varepsilon)\right]=-\omega^2\mu\varepsilon+j\omega\mu\sigma=-\omega^2\mu\varepsilon\left(1-j\frac{\sigma}{\omega\varepsilon}\right)$$ Useful later

If $\mathbf{J}=\frac{\partial\mathbf{D}}{\partial t}=(\omega\varepsilon''-j\omega\varepsilon')\mathbf{E}_1\,\mathrm{e}^{i\omega t}$

Let $\mathbf{E}=\mathbf{E_1}e^{i\omega t}$ in a capacitor, then:

$\frac{1}{2}\mathbf{E}\cdot\mathbf{J}^{*}=\frac{\omega\varepsilon''E_1^2}{2}$ = power lost per unit volume of a material

$\frac{\varepsilon'E^2}{2}=\frac{\varepsilon'E_1^2}{2}$ Maximum stored energy density (Wang – Solid State Electronics pg 415)

$$Q=\frac{maximum\ stored\ energy}{average\ power\ loss}\cdot\omega=\frac{\varepsilon'}{\varepsilon''}=\frac{1}{\tan\delta_a}$$

Q is the quality factor of the capacitive material = 1/(loss tangent)